Car speed estimation

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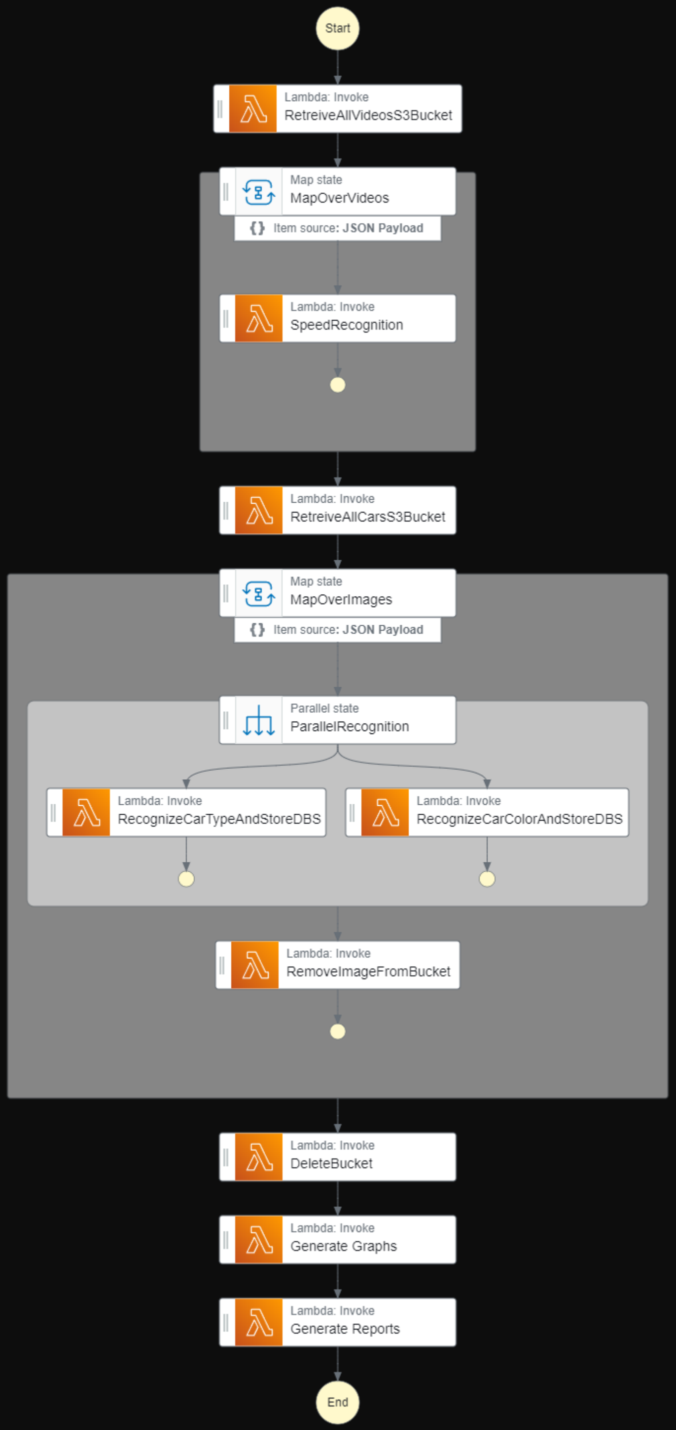
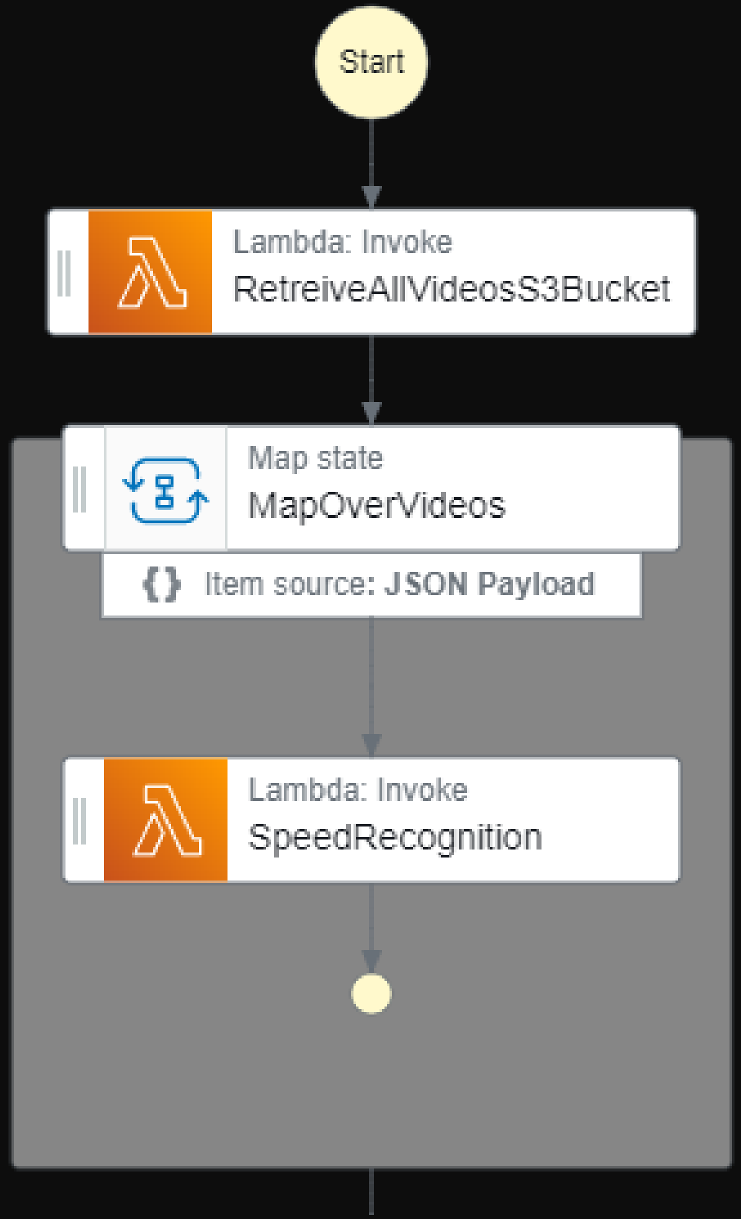
Motivation

- Car speed estimation
- Detection of speeding cars
- Creating Statistics

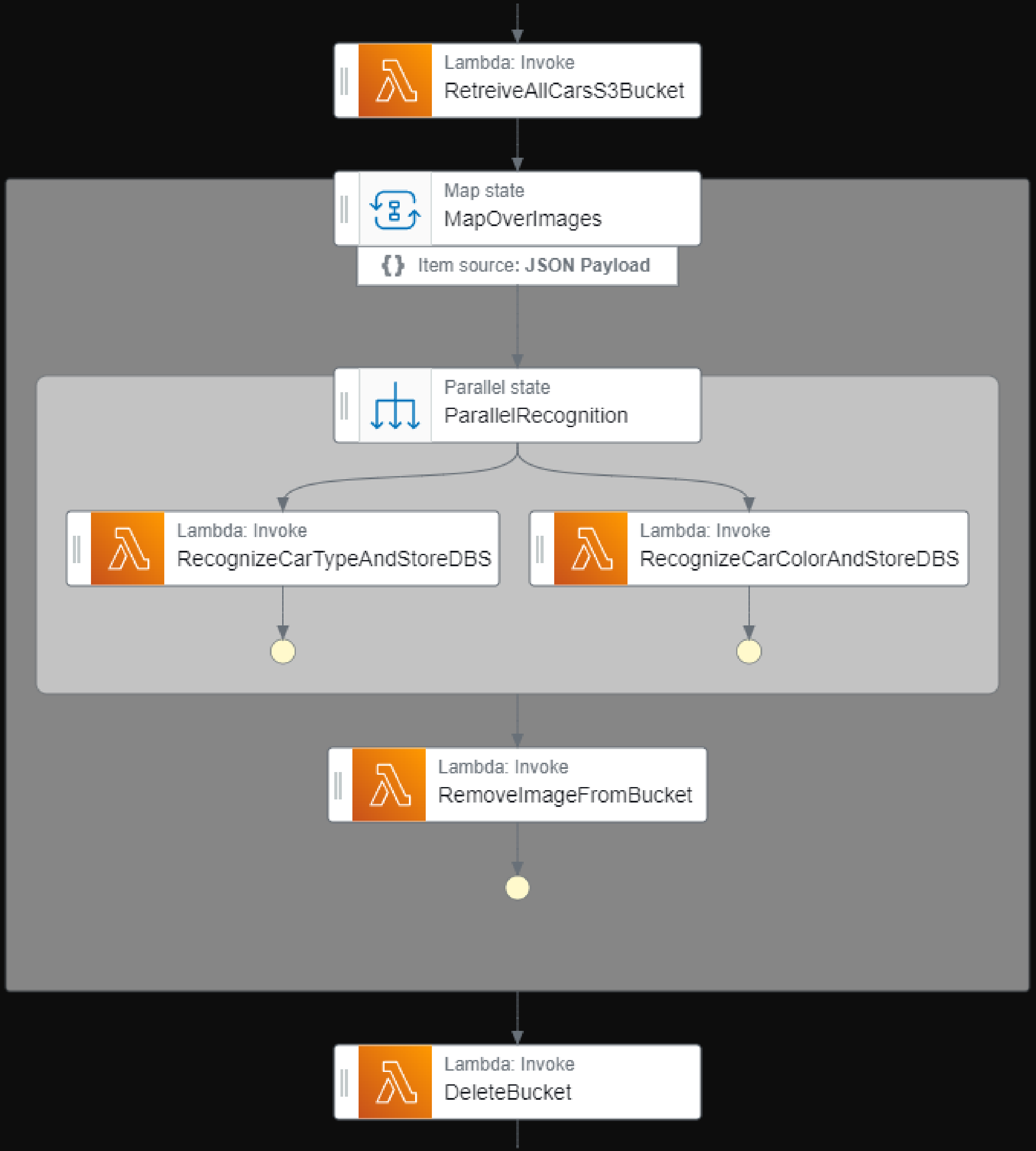

Start
Lambda: Invoke
RetreiveAllVideosS3Bucket
Map state
MapOverVideos
{} Item source: JSON Payload
Lambda: Invoke
SpeedRecognition
Lambda: Invoke
RetreiveAllCarsS3Bucket
Map state
MapOverImages
{} Item source: JSON Payload
Parallel state
ParallelRecognition
Lambda: Invoke
RecognizeCarTypeAndStoreDBS
Lambda: Invoke
RecognizeCarColorAndStoreDBS
Lambda: Invoke
RemoveImageFromBucket
Lambda: Invoke
DeleteBucket
Lambda: Invoke
Generate Graphs
Lambda: Invoke
Generate Reports
End

Workflow



Workflow

- Loop over all livestreams/videos
- Recognize cars and save screenshots
- Estimate the speed between configured lines

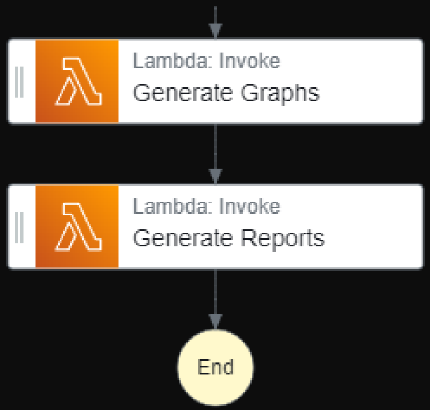


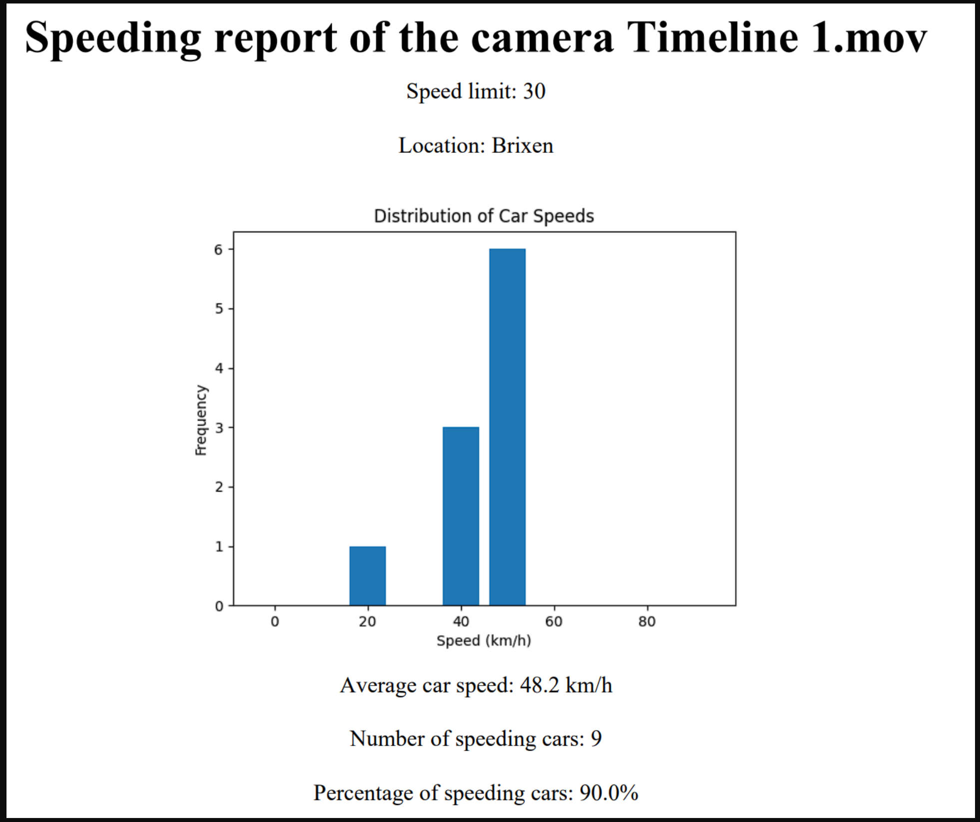
Workflow

- Loop over all collected car images
- Recognize Type and Color of cars
- Clean up

Workflow

- Create Statistics per Camera




Speeding report of the camera Timeline 1.mov

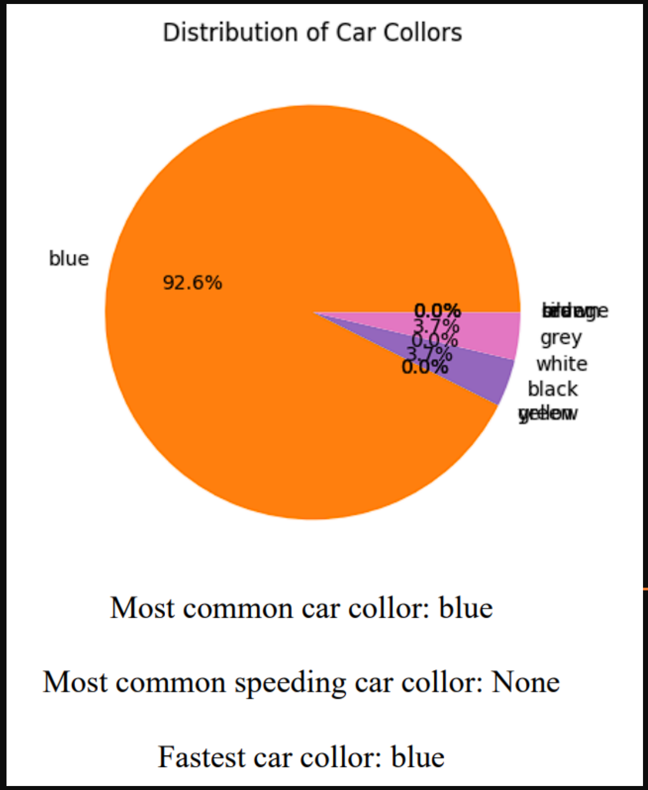
Speed limit: 30

Location: Brixen

Average car speed: 48.2 km/h

Number of speeding cars: 9

Percentage of speeding cars: 90.0%



Most common car collor: blue

Most common speeding car collor: None

Fastest car collor: blue

Live Demo

Evaluation

- Lambda SpeedRecognition:
  - Video 1:
    - Video duration: 1:04 min
    - Execution duration: 2:31 min
    - Allocated Memory: 10240 MB
    - Cost:
      - 0.02€ per min
      - 1.38€ per hour
  - Video 1:
    - Video duration: 1:04 min
    - Execution duration: 4:30 min
    - Allocated Memory: 4096 MB
    - Cost:
      - 0.017€ per min
      - 1.02€ per hour

- Lambda CarTypeRecognition and CarColorRecognition
  - Ca. 15 Cars per min:
    - Car Type Recognition: 0.001€ per function call
    - Car Color Recognition: 0.00075€ per function call
    - Total: 0.01125€

Conclusion

- Problems integrating large libraries in lambda
- AWS implementation of car speed recognition not available anymore
- Better implementation using EC2 with GPU

Thank you for your Attention